

T LOHITH

Cybersecurity Engineer • OT/ICS & Blue Team Security • SOC Analyst

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SUMMARY

M.Tech Cybersecurity candidate specializing in OT/ICS security and Blue Team operations — including a live-deployed Modbus TCP honeypot (13 MITRE ATT&CK detection rules, real attacker data on OVHcloud VPS) and a production SIEM with ML anomaly detection across 3 log sources. Seeking SOC Analyst/Security Engineer roles to detect, triage, and eliminate real threats in critical infrastructure.

EDUCATION

M.Tech — Networks & Cybersecurity

Aug 2024 – June 2026 (Expected)

Amity University Gurugram • Thesis: OT/ICS Modbus TCP Honeypot with Graduated Response & MITRE ATT&CK Detection

B.Tech — Cybersecurity, IoT & Blockchain

Aug 2019 – May 2024

Siddharth Institute of Engineering & Technology, Tirupati

EXPERIENCE

Ethical Hacking Intern • InLighnX Global Pvt. Ltd. • Remote

Jul–Aug 2025

- Developed Python keylogger (keystroke capture, screenshot exfiltration, registry persistence, .exe packaging) simulating red team malware; performed command injection attacks on DVWA/bWAPP with reverse shell execution via Netcat.

Full Stack Python Developer Intern • Aavanto Pvt. Ltd. • Remote

Oct–Dec 2023

- Built data visualization modules for smart grid analytics platform using Python and Flask; contributed to IoT energy automation modules.

PROJECTS

OT/ICS Modbus TCP Honeypot — Live Deployed — Python • PyModbus • Flask • Docker • OVHcloud • MITRE ATT&CK for ICS

- Live on public VPS (Port 502, Ubuntu 22.04) — Shodan-indexed, actively collecting real attacker sessions. 3 virtual PLCs (conveyor, hydraulic pump, heater) with physics-based register simulation (temp/pressure/RPM); graduated response system (Monitor→Warn→Throttle→Block) and phantom writes extend attacker engagement.
- 13 MITRE ATT&CK for ICS detection rules (T0836, T0843, T0846, T0855, T0862); real-time Flask dashboard with geo-mapped IPs, PLC topology, MITRE technique charts, and per-attacker threat scoring.

OT/ICS SIEM Dashboard — Python • Flask • SQLite • Chart.js • Leaflet.js

- Production SIEM ingesting SSH, Apache/Nginx, and SCADA logs with correlation rules (brute force, dir scanning, admin probing), MITRE ATT&CK mapper, AbuseIPDB/VirusTotal threat intel, Isolation Forest anomaly detection, and automated IR playbook generator.

Malware Analysis Lab — FlareVM • REMnux • PESTudio • Wireshark • INetSim • YARA

- Static (PEiD, PESTudio, Strings) + dynamic (ProcMon, Process Hacker, Wireshark + INetSim) analysis of real malware samples from MalwareBazaar; authored YARA detection rules per sample with MITRE ATT&CK TTP mapping and IOC reports.

Network Traffic Analyzer — OT/ICS Protocol Detection Python • Scapy • Flask • SQLite • Chart.js

- Real-time packet capture engine (thread-safe, queue-based, BPF filter support) with OT protocol fingerprinting for Modbus TCP, DNP3, and IEC 104; thread-safe SQLite backend (WAL mode) with flow tracking, top-talkers aggregation, and protocol distribution — optimized with UPSERT and indexed queries.

Threat Intelligence Aggregator — Python • AbuseIPDB • VirusTotal • Shodan • AlienVault OTX • Flask

- Multi-source threat intel pipeline aggregating IP reputation, domain analysis, and IOC enrichment from AbuseIPDB, VirusTotal, Shodan, and AlienVault OTX; unified scoring engine with REST API output and exportable threat reports for SOC analyst consumption.

Web Vulnerability Scanner — Python • asyncio • OWASP Top 10 • Requests • BeautifulSoup

- Async Python scanner testing OWASP Top 10 vulnerabilities including SQLi, XSS, open redirects, insecure headers, and directory traversal; generates structured vulnerability reports with severity ratings, proof-of-concept payloads, and remediation guidance.

Advanced Linux Log Analyzer — SSH Intrusion Detection & Auto-Response — Python • ipset • iptables • SMTP • Linux • auth.log

- Real-time SSH brute-force detection engine monitoring /var/log/auth.log with configurable failure threshold, sliding time window, and deduplication; automatically blocks offending IPs via ipset/iptables with timed cooldown and auto-unblock, and dispatches SMTP email alerts to administrators on threshold breach.

TECHNICAL SKILLS

SOC / SIEM

Splunk (in progress), Microsoft Sentinel, ELK Stack, custom SIEM, alert triage, log analysis, incident response, SOAR concepts

OT / ICS

Modbus TCP, DNP3, IEC 104, SCADA architecture, PyModbus, ICS honeypot design, MITRE ATT&CK for ICS

Malware

FlareVM, REMnux, PESTudio, PEiD, ProcMon, Process Hacker, INetSim, Wireshark, YARA rule writing, Cuckoo Sandbox

Network Sec.

Scapy, Nmap, Snort/Suricata, IDS/IPS deployment, VLAN segmentation, routing security, 802.11 wireless (WPA2/WPA3, rogue AP detection)

Offensive

Burp Suite, Metasploit, Kali Linux, DVWA, bWAPP, command injection, reverse shells

Dev & Tools

Python (advanced), C(Basic), Flask, Docker, Git/GitHub, SQLite, Linux, REST APIs, MITRE ATT&CK, NIST CSF, OWASP Top 10

CERTIFICATIONS

Google Cybersecurity Professional Certificate — 9 courses: SIEM, IDS, Python, Linux, threat analysis • Coursera / Google Feb 2026

CompTIA Security+ - In Progress • TryHackMe Pro (SOC L1, Blue Team) • HackerArise: SCADA Security, Web App Hacking, OSINT2026